Niagara 4 point deployment · Part 1 of 2

Publishing Station Points

The station as a server: BACnet/IP, oBIX, Modbus TCP slave

Verified on Niagara 4.14.0.162 · every object read back by an independent client

Part 2 covers the opposite direction: building clients that collect these points

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One pattern, three protocols

Publishing a point is never one operation the step that fails is the second

Step	BACnet export	oBIX export	Modbus TCP slave
1 · Create the object	REST add□bacnet:*Descriptor	REST add□obixDriver:ObixExport	REST cp□(add leaves NullProxyExt)
2 · Bind source / address	pointOrd oBIX PUT	point oBIX PUT	dataAddress REST mod
3 · Public identity	objectId REST mod	binding is the identity	binding is the identity
Channel count	REST + oBIX	REST + oBIX	REST only

Why step 2 is the one that breaks

A source binding is a BOrd slot. The REST encoder cannot express that type: it answers BString cannot be cast to BOrd, and the slot stays null.

A Modbus slave point created with add carries a NullProxyExt and has no dataAddress slot at all there is nothing to configure afterwards.

A BACnet descriptor is valid and still reads {fault} until objectId is assigned; a null source point faults for a second, unrelated reason.

Where the objects live

Three protocols, three trees and only one of them is a single flat table

BACnet/IP

```
/Drivers/BacnetNetwork
  bacnetComm/network/ipPort
    link  (adapter    )
  localDevice
    exportTable
      <descriptor>     1 point
```

oBIX

```
/Drivers/ObixNetwork
  server   (servlet)
  exports
    <export>     1 point

same /obix/config tree
```

Modbus TCP slave

```
/Drivers/<slaveNetwork>
  port  (ServerPort)
  <slaveDevice>
    points
      <point>
        dataAddress
```

BACnet is the only protocol with a separate network-enable layer above the objects: the network, the port and the link must all be up before a descriptor can answer on the wire. oBIX and the Modbus slave listen as soon as their own object is enabled.

What the trees imply for a build order

BACnet: enable top-down. A descriptor added to a disabled network is accepted and answers nothing.

oBIX: the servlet and the export folder are siblings one is the door, the other is the list, and neither depends on the other being edited.

Modbus slave: the port belongs to the network, so one port serves every slave device under it.

BACnet/IP: three enable layers before anything answers

Enabling only the port leaves it {ok} while nothing replies on UDP 47808

Layer	Path	Failure when off
Driver network	/Drivers/BacnetNetwork	whole network {disabled}, every child inert
Network port	/bacnetComm/network/ipPort	{disabled,fault} on that component
Link	/ipPort/link	BACnet/IP adapter not ready! Check cable connections.

The link holds three enum slots

adapter, adapterId and ipAddress are DynamicEnum slots. REST rejects them twice BEnumRange cannot be cast to BEnum for the value, Illegal name for the display text. The one working channel is an oBIX PUT of the escaped name read from the slot's ~range.

```
GET  /ipPort/link/adapter/~range
PUT  /ipPort/link/adapter/  <enum val="Intel$28R$29$20PRO$2f1000$20MT$20Desktop$20Adapter$20$232"/>
```

adapterId and ipAddress follow automatically (eth6 / the station's own address)

> A virtual or debug adapter is a common wrong value and produces adapter not ready. Completion is four green states at once: network {ok}, ipPort {ok}, empty faultCause, localDevice Operational.

BACnet export table: one descriptor per point

Three slots, two channels and a component that renames itself

Slot	Type	Channel	Rule
objectName	String	REST mod	unique on the whole network
pointOrd	BOrd	oBIX PUT only	one source point backs one descriptor
objectId	BBacnetObjectIdentifier	REST mod	e.g. analogValue:100

Descriptor type follows the source point

Source	Descriptor
control:NumericPoint (read-only)	BacnetAnalogInputDescriptor
Numeric, value / output semantics	BacnetAnalogValue / AnalogOutputDescriptor
control:BooleanPoint	BacnetBinaryInputDescriptor
control:EnumPoint	BacnetEnumDescriptor

Writing objectId renames the component: demo_av100 becomes analogValue_100. Look it up by listing the table, never by the name you typed.

A # in the source ord must be escaped as $23. A second descriptor on the same source point faults.

pointOrd can answer BadUriErr and still have taken effect confirm by reading the slot back, not by the response code.

oBIX export: the binding only oBIX can write

The same BOrd limit, seen from the other side

Create over REST, bind over oBIX

add obixDriver:ObixExport not obix:ObixExport, which answers type-not-found then PUT the source ord into the point slot.

```
PUT  /obix/config/Drivers/ObixNetwork/exports/<name>/point/
<uri val="station:|slot:/<path>/<point>"/>    #  # in the path becomes $23
```

Writing through an export

POST to the export's set node, not to the export object. A proxied source point accepts the write and does not move that is a mapping limit, not an authentication problem.

```
POST /obix/config/Drivers/ObixNetwork/exports/<name>/set/   <real val="25.5"/>
```

Attempt	Result
Create a component (POST type, PUT child path)	BadUriErr every variant
Add a history or alarm extension	BadUriErr
Write a proxied point's value	set node is inert

So oBIX owns three things in this workflow: binding BOrd slots, writing DynamicEnum names, and reading history and alarms. Creating the object belongs to REST.

Modbus TCP slave: copy a point, do not add one

The created object looks like a point and publishes nothing

	Slave (server)	Client (other direction)
Network	modbusTcpSlave:ModbusTcpSlaveNetwork	modbusTcp:ModbusTcpNetwork
Device	modbusTcpSlave:ModbusSlaveDevice	modbusTcp:ModbusTcpDevice
Proxy ext	ModbusServerNumericProxyExt	ModbusClientNumericProxyExt
Address slot	dataAddress (FlexAddress)	dataAddress (FlexAddress)
Who initiates	external master	station
Polling	none no pollScheduler	devicePollConfig + pollScheduler

Two failures worth recognising

add produces a point with a NullProxyExt and no dataAddress slot. cp from a point that already carries ModbusServerNumericProxyExt is the working route.

Property 'address' not found means the slot name is wrong, not that the point is broken read the real slot list before guessing.

A slave network has no pollScheduler because it answers rather than polls. That absence is normal, not a fault.

> An external write enters the station through the proxy and lands in the priority array. If the point is also driven internally, the write can be masked check out for {overridden} before concluding the register was ignored.

The station's own read-back is not proof

Each protocol has one outside call that settles it

Protocol	Outside call	What it proves
BACnet/IP	unicast Who-Is to UDP 47808, then ReadProperty 85	the server answers and the object carries the live value
oBIX	GET /obix/config/ with Basic auth	the export is reachable and typed
Modbus TCP slave	FC03 read from a master off-station	the register maps to the point you think it does

BACnet Who-Is, minimal frame

```
pkt = bvlc(01 00 10 08)        # unicast to <HOST>:47808, not a global broadcast
response  810a 0014 0100 1000 c402 0004 5822 05c4 9100 2124
        device 1112 · maxAPDU 1476 · vendor 36 (Tridium)
```

A global broadcast Who-Is sent from the same host is frequently dropped. Use unicast to the station's own address.

For ReadProperty, objtype 2 is analogValue and prop 85 is present-value; prop 77 is object-name when you need to confirm the identity.

The device instance in the reply is the station's own not the field device's, which is the Part 2 direction.

Fault reference and what stays open

Every row below was observed on a live station, not inferred

Symptom	Cause	Fix
BACnet/IP adapter not ready!	link bound to a virtual or debug NIC	rebind adapter by escaped name
Invalid value for BACnet Object:- {null}	descriptor bound to a point with no value	write a value to the source
Duplicate Object Name	objectName empty or reused	assign a unique name
Cannot find exported point	wrong ord, or # unescaped	rewrite pointOrd, # $23
descriptor not found by name	it renamed itself after objectId	list the export table
point has NullProxyExt, no dataAddress	created with add	cp from a real slave point
Property 'address' not found	wrong slot name	use dataAddress
export reads {fault}	oBIX object type wrong	add obixDriver:ObixExport

Open, and stated as open

Clearing a priority slot to true null is not offered by either channel; both leave the slot occupied with a NaN or an override. Design so a slot never needs emptying.

SNMP agent exports were built but not verified end to end from outside, and are therefore left out of this part.